LECTURE-3

Rules for finding Integrating Factor

Rule-6:

Theorem

If $mn_1 - m_1 n \neq 0$ *then* $x^{\alpha} y^{\beta}(mydx + nxdy) + x^{\alpha_1} y^{\beta_1}(m_1 ydx + n_1 xdy) = 0$ *has an integrating factor of the form* $x^h y^k$*, where* h, k *are given by*

$$\frac{h+1+\alpha}{m} = \frac{k+1+\beta}{n} \quad \& \quad \frac{h+1+\alpha_1}{m_1} = \frac{k+1+\beta_1}{n_1}.$$

Example 4: $(xy^2 + 2x^2y^3)dx + (x^2y - x^3y^2)dy = 0.$

Solution

Let $x^h y^k$ *be an I.F. of the given differential equation. Then form the condition of exactness* $M_y = N_x$ *we have* $h = -3, k = -3.$

$\therefore F = \int \left(1/(x^2y) + 2/x\right) dx + \psi(y) = -\frac{1}{xy} + 2\log x + \psi(y).$

Since $F_y = N$*, so we have* $\psi'(y) = -1/y \Rightarrow \quad \psi(y) = -\log y +$ *constant.*

$\therefore$ *The desire solution is* $\log \frac{x^2}{y} = C + \frac{1}{xy}$.

Exercise 2 : $x(3ydx + 2xdy) + 8y^4(ydx + 3xdy) = 0$. [h=1, k=1].

Exercise 3: $(8y + 5xy^2)dx + (6x + 4x^2y)dy = 0$. [h=3, k=2]

Rules for finding Integrating Factor

Rule-7:

Theorem

The differential equation $M(x,y)dx + N(x,y)dy = 0$ has an integrating factor $u(t)$, which is a function solely of a single argument $t(x,y)$, if and only if $\frac{M_y - N_x}{Nt_x - Mt_y}$ is a function of t only, say $\phi(t)$, and in this case $u(t)$ must be a constant multiple of $e^{\int \phi(t)}dt$.

We consider the following cases in particular.

1. For $t = x + y$, $\frac{M_y - N_x}{N - M}$ should be a function of $x + y$ alone.
2. For $t = x^2 + y^2$, $\frac{M_y - N_x}{2(Nx - My)}$ should be a function of $x^2 + y^2$ alone.
3. For $t = x^n + y^n$, $\frac{M_y - N_x}{n(Nx^{n-1} - My^{n-1})}$ should be a function of $x^n + y^n$ alone.
4. For $t = xy$, $\frac{M_y - N_x}{(Ny - Mx)}$ should be a function of xy alone.
5. For $t = \frac{y}{x}$, $\frac{-x^2(M_y - N_x)}{(Ny + Mx)}$ should be a function of y/x alone.

Example 5: $(3y^2 - x)\,dx + 2y\,(y^2 - 3x)\,dy = 0.$

Let $t = x + y^2$. Then $t_x = 1$ and $t_y = 2y$. Let $M = 3y^2 - x$ and $N = 2y(y^2 - 3x)$. Then $M_y = 6y$ and $N_x = -6y$.

$\therefore Nt_x - Mt_y = -4y^3 - 4yx$. Hence, $\frac{M_y - N_x}{Nt_x - Mt_y} = -\frac{3}{y^2+x} = -\frac{3}{t}$.
$\therefore e^{\int \phi(t)dt} = e^{-\int \frac{3}{t}dt} = \frac{1}{t^3}$.

$$\begin{aligned}
F(x,y) &= \int \frac{3y^2 - x}{(x+y^2)^3}dx + \psi(y) = \int \frac{4y^2 - (x+y^2)}{(x+y^2)^3}dx + \psi(y) \\
&= \frac{x - y^2}{(x+y^2)^2} + \psi(y). \\
\Rightarrow \quad F_y(x,y) &= \frac{2y(y^2 - 3x)}{(x+y^2)^3} + \psi'(y) = \frac{N}{t^3} + \psi'(y) \\
\Rightarrow \quad \psi'(y) &= 0 \quad \Rightarrow \quad \psi(y) = c_1.
\end{aligned}$$

Therefore, the solution is $\frac{x - y^2}{(x+y^2)^2} = C$.

Example 6: $[x(x^2+y^2)^2-y]dx+[x+y(x^2+y^2)^2]dy=0.$

Let $t=x^2+y^2$, then $\dfrac{M_y-N_x}{2(Nx-My)}=\dfrac{-1}{x^2+y^2}=\dfrac{-1}{t}$.

So, the I.F. is $e^{\int(1/t)dt}=\dfrac{1}{t}=\dfrac{1}{x^2+y^2}$.

$\therefore$ The solution is $(x^2+y^2)^2+4\tan^{-1}(\dfrac{y}{x})=c$.

Linear Equation of First Order

$$\frac{dy}{dx} + P(x)y = Q(x) \tag{4.1}$$

Rewriting eq. (4.1),

$$(Q(x) - P(x)y)\ dx - dy = 0$$

Let $M(x, y) = Q(x) - P(x)y$ & $N(x, y) = -1$. $\frac{1}{N}(M_y - N_x) = P(x)$.

Thus, the integrating factor $(I(x))$ according to Theorem 4 is $e^{\int P(x)\,dx}$.

$$\begin{aligned}
\therefore\ e^{\int P(x)\,dx}(Q(x) - P(x)y)\ dx - e^{\int P(x)\,dx}dy &= 0 \\
\Rightarrow \quad \frac{d}{dx}\left(ye^{\int P(x)\,dx}\right) &= Q(x)e^{\int P(x)\,dx} \\
\Rightarrow \quad ye^{\int P(x)\,dx} &= \int Q(x)e^{\int P(x)\,dx}\,dx + c
\end{aligned}$$

Thus, $y = e^{-\int P(x)\,dx}\left(\int Q(x)e^{\int P(x)\,dx}\,dx + c\right)$.

Example: $xy' + (1-x)y = e^{2x}$.

$$y' + \frac{1-x}{x}y = \frac{e^{2x}}{x}$$

$P(x) = \dfrac{1-x}{x}$ and $Q(x) = \dfrac{e^{2x}}{x}$. Then

$$\text{I.F. } I(x) = e^{\int P(x)\,dx} = e^{\int \frac{1-x}{x}\,dx} = xe^{-x}$$

$$\begin{aligned} yxe^{-x} &= \int \frac{e^{2x}}{x} xe^{-x}\,dx + c = e^{x} + c \\ \Rightarrow y &= \frac{e^{2x}}{x} + \frac{ce^{x}}{x}. \end{aligned}$$

Example: A body of mass m is dropped from rest from height h_0 in the earth's atmosphere. Assuming that the body falls vertically under the action of earth's gravity and neglecting all other effects except air resistance, which is proportional to its velocity, find its height h at any subsequent time.

Bernoulli's Equation

$$\frac{dy}{dx} + P(x)y = Q(x)y^n \tag{5.1}$$

Let $v = y^{-n+1}$ for $n \neq 1$. Then $\frac{dv}{dx} = (-n+1)y^{-n}\frac{dy}{dx}$.

Eq. (5.1) gives

$$\frac{dv}{dx} + (-n+1)P(x)v = (-n+1)Q(x)$$

$$\begin{aligned} v(x) &= e^{(n-1)\int P(x)\,dx}\left[(1-n)\int Q(x)e^{(1-n)\int P(x)\,dx} + c\right] \\ \Rightarrow \quad y^{1-n} &= e^{(n-1)\int P(x)\,dx}\left[(1-n)\int Q(x)e^{(1-n)\int P(x)\,dx} + c\right] \end{aligned}$$